# Gayathri Krishnamoorthy

## Electrical Engineer| Computer Scientist

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Power Systems Engineer with 3+ years of experience as a researcher at Washington State University and Internship experience at Pacific Northwest National Laboratory (PNNL). Worked on multiple projects developing software tools to efficiently integrate renewable resources into power grid models. I am outgoing, creative, and detail-oriented with strong engineering basics and communication skills. I am effective both solo or on a team. In my spare time I practice yoga, meditation, and painting. I am also passionate about the brain sciences behind mindfulness and keeping up with recent developments in AI.

## Education

**Present – August 2018** | **Ph.D. in Electrical Engineering , Washington State University, Pullman, WA**

- Research : Transmission Distribution Co-simulation : Utility of Distribution Connected Renewable Resources in Providing Transmission System Support
- Major : Power Systems
- Minor : Computer Science
- Advisor : Dr. Anamika Dubey

**May 2018 – Aug 2016** | **M.S. in Electrical Engineering, Washington State University, Pullman, WA**

- Research : An Iterative Co-simulation Framework for the Integrated Transmission and Distribution System Analysis
- Major : Power Systems
- Minor : Computer Engineering
- Advisor : Dr. Anamika Dubey

**May 2016 – Aug 2012** | **B.E. in Electronics and Communications Engineering, Anna University, Chennai, India**

- Title : Smart Home Design with Voice Recognition in a Single Controller (Arduino Uno) Environment
- Major : Embedded Systems
- Minor : Communications
- Advisor : Dr. Suresh T

## Professional Experience

**Present – August 2016** | **Graduate Research Assistant, Washington State University, Pullman, WA**

- Designed and developed a transmission and distribution (T&D) co-simulation platform to understand the impacts of increasing renewable penetrations on power system operations and evaluated various power quality issues.
- Developed analytical models to demonstrate the convergence of T&D co simulation platform.
- Developed a hybrid T&D co-simulation platform that assists in secondary frequency regulation services to the power grid using distribution connected batteries.
- Developed an optimization algorithm in a decentralized distribution market operation scenario to optimize the proposed market bids without violating system operating constraints.

MATLAB | OpenDSS | GridPV | Python | Power World

Dec 2019
Aug 2019

**Graduate Teaching Assistant, Washington State University, Pullman, WA**
Course : Numerical Computing for Engineers
Tasks :
- Engaged a class of 80 undergraduate (multiple engineering discipline) students
- Helped the class in learning to use MATLAB, a professional computing software)
- Provided my best efforts in evaluating students' homeworks and exams

MATLAB

May 2019
Jan 2019

**Graduate Teaching Assistant, Washington State University, Pullman, WA**
Course : Professional Skills in Computing and Engineering
Tasks :
- Engaged a class of 108 undergraduate electrical and computer engineering students
- Taught the class to use Microsoft project professional software
- Detailed lectures on professional writing using LaTex.

Microsoft Project Professional | LaTex

Dec 2018
May 2018

**Graduate Research Intern, Pacific Northwest National Laboratory, Seattle, WA**
- Assisted in development of a transactive energy software agent, TESP, that optimizes the dynamics of distributed energy resources in a household and bids the excess energy in the energy market to provide monetary benefits to the costumers using electric vehicles, batteries, and PVs.
- Performed scalability testing for the open source co-simulation platform (HELICS) developed at PNNL in multiple programming language environments.
- Performed testing of the transactive energy simulation platform with HELICS that integrated multiple power system market components on a Linux environment
- Developed a C++ wrapper for MATPOWER software
- Developed a Docker Engine for the FNCS co-simulation software platform.

C++ | Python | MATLAB | MATPOWER | GridLAB-D | OpenDSS | Docker | EnergyPlus | FNCS | TESP | HELICS

Dec 2014

**In-Plant Training, HCL Technologies, Chennai, India**
- Developed fundamental principles and knowledge of recent advancements in Embedded Systems, Cloud Computing, Internet of Things, Android, and .Net software platforms.
- Attended robotics and embedded systems workshop on making of legged robots and successfully completed the project.
- Developed a prototype racing car that used Freescale controllers to automatically recognize the specially designed racetrack for the Freescale Cup intelligent car racing competition.

HTML | C | C++ | Python | Arduino | Raspberry Pi | EasyVR

## Professional Activities

| | |
|---|---|
| Present | Serving as a reviewer for IEEE Power and Energy Society Conferences, Institution of Engineering and Technology Journals, and IEEE Systems Journal |
| 2019 | Authored a Ph.D. Research Proposal to Pacific Northwest National Laboratory |
| 2018 | Paper Presentation- IEEE Power and Energy Society General Meeting |
| 2017 | Poster Presentation - Power Systems Engineering Research Center |
| 2015 | Attended Internet of Things with Arduino workshop for real-time applications at AltSense |
| 2015 | Trained and participated in FreescaleCup Intelligent Car Racing Competition |

## Programming Languages

| | |
|---|---|
| C++ | ●●●●● |
| MATLAB | ●●●●● |
| C | ●●●●● |
| Python | ●●●●○ |
| HTML | ●●●●○ |
| Javascript | ●●●○○ |
| CSS | ●●●○○ |

## Simulation Tools

- OpenDSS
- GridLAB-D
- Simulink
- MATPOWER
- EnergyPlus
- FPGA Design
- Raspberry Pi
- Arduino Uno

## Projects

### Real-time Control of Energy Storage Sytems for Frequency Regulation of Power Grid — 2019

Energy Market

In this project, I am looking to explore the utility of DER's in transmission system support especially, frequency regulation. There are several aspects to this project : 1) Learning consumer willingness and flexibility in providing grid services using deep learning techniques and 2) Energy market structure aggregation from the customer level to the transmission operators level.

MATLAB | OpenDSS | Python

### GMLC-TDC/ HELICS — 2018

github.com/GMLC-TDC  github.com/GMLC-TDC/MATPOWER-wrapper  HELICS

HELICS is a coordinated effort by multiple National Labs to develop a platform that aids advanced power system operations. I worked on certain aspects of the development like scalability testing and wrapper development for multiple power system tools. One major takeaway during this internship was learning docker development.

C++ | Python | MATLAB | Matpower

### TESP — 2018

github.com/pnnl/tesp  TESP

For this project, I was involved in developing the energy market structure (real-time and day-ahead) for HVAC (heating, ventilation and air conditioning) systems. The major tasks were to develop consensual bidding strategies and test/validate on large scale power system networks.

Python | C++ | GridLAB-D | EnergyPlus

### Transmission -Distribution Co-simulation — 2017 - 2019

github.com/yvelaga/PSERC-Cosimulation  PSERC T-60

Developed a framework for the integrated T&D system analysis to understand the impacts of distribution-connected renewable resources (DERs) on transmission system operations and vice-versa (project funded by PSERC). Also, using the co-simulation framework developed, explored the utility of DER's in transmission system support.

MATLAB | OpenDSS | Python

### Website for Distribution lab at WSU — 2019

DOPAL

Designed and maintain a professional website for my research group at WSU.

HTML | bootstrap | CSS | jquery

### Real Time Cyber-Physical System Testbed Development — 2017

Course Project

Developed a real time cyber-physical system testbed communicating with each other through MatrikonOPC software to analyze the impacts of a real-time cyberattack on the power grid. Designed an Intrusion Detection System (IDS) in substations using SNORT software to provide additional security along with firewalls.

Matrikon OPC | MATLAB | Simulink | C++

### Multi-core Processor Design — 2016

github.com/adititihuse/Branch-Target-Buffer-Simulation

Development of an advanced multi-core processor design in C++ that handles functional units looping instructions for static and dynamic branch predictions

C++

## Honors and Awards

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|---|---|
| 2019 | **Awardee** Graduate & Professional Student Association Scholarship Recipient for the Research Exposition at WSU |
| 2018 | **2nd Place** Best Student Paper Award at North American Power Symposium |
| 2015 | **Awardee** Completed course in CCNA Exploration : Network Fundamentals |
| 2015 | **2nd Place** Texas Instruments India Analog Maker Competition |
| 2014 | **Awardee** Level 1 Cambridge English Language Assessment in Business English |
| 2011-2012 | **Scholarship** 100 % fee concession in high school |

## Publications

2019 G. Krishnamoorthy and A. Dubey, ”Transmission–Distribution Cosimulation : Analytical Methods for Iterative Coupling”, in IEEE Systems Journal.

2019 J. C. Bedoya, C. Liu, G. Krishnamoorthy and A. Dubey, ”Bilateral Electricity Market in a Distribution System Environment”, in IEEE Transactions on Smart Grid.

2019 Sen, P.K., Velaga, Y.N., Chen, A., Krishnamoorthy, G. and Dubey, A., 2019. Advancements in Co-Simulation Techniques in Combined Transmission & Distribution Systems Analysis. The Journal of Engineering.

2019 G. Krishnamoorthy, A. Dubey and P. K. Sen, ”Iteratively-Coupled Co-simulation Framework for Unbalanced Transmission-Distribution System,” 2019 IEEE Milan PowerTech, Milan, Italy, 2019, pp. 1-6.

2018 Krishnamoorthy, Gayathri, and Anamika Dubey. ”A framework to analyze interactions between transmission and distribution systems.” In 2018 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2018.

2018 Y. N. Velaga, A. Chen, P. K. Sen, G. Krishnamoorthy and A. Dubey, ”Transmission-Distribution Co-Simulation : Model Validation with Standalone Simulation,” 2018 North American Power Symposium (NAPS), Fargo, ND, 2018, pp. 1-6.

## References


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